BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LANs, WANs and Network Models

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Convert 00101101 binary to denary.

2. What is the Internet?
   A) A network of networks
   B) One central web server
   C) Only public web pages
   D) One school's wireless LAN

3. "A web client must host every website before it can request a page." Correct this. Explain why.

4. A pupil needs work on several devices. "Cloud files are always available when the Internet fails." Correct this. What could happen?

Today's objectives

Explain the core ideas of lans, wans and network models using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

LAN
Covers a small geographical area such as one building, usually under a single organisation.

WAN
Connects networks over wider geographical areas, often via third-party infrastructure.

Client-server
Central servers provide resources/services for client devices.

Peer-to-peer
Devices share resources directly without a dedicated central server for those resources.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

LAN
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I Do - Worked model

A company has two offices in different cities connected so staff share files. Identify network types.

- Each office can use its own LAN.
- A WAN connects geographically separate LANs.
- A server can centrally control shared files and access.
- Explain why peer-to-peer may be less suitable for many managed users.

We Do - Guided practice

1. Compare a five-computer home network with a national retailer's network.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write the missing output for a connection between two distant school sites.

```
separateSites = true
if separateSites then
...
else
print("One LAN")
endif
```

You Do - Independent exam questions

Q2. Explain one difference between LAN and WAN. [2 marks]

Q3. Give one advantage of client-server networking. [2 marks]

Q4. Explain one drawback of peer-to-peer for a large school. [3 marks]

Exit Ticket - True or False?

A WAN can connect LANs in geographically separate locations.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. 45, because 32 + 8 + 4 + 1 = 45.

2. A - A network of networks

3. A server stores and makes website resources available for clients to request.

4. Cloud resources are accessed over a network. Files may be unavailable if the connection or server fails.

I Do - Explain the model

Each office can use its own LAN.

A WAN connects geographically separate LANs.

A server can centrally control shared files and access.

Explain why peer-to-peer may be less suitable for many managed users.

We Do - Model answers

1. Home is LAN; retailer uses LANs connected by WAN.

Code First - Answer

```
separateSites = true
if separateSites then
print("WAN connects sites")
else
print("One LAN")
endif
```

Expected output: WAN connects sites

You Do - Self-mark

Q2: LAN covers smaller/local area; WAN connects over wide geographical areas.

Q3: Centralised user account, access or resource management.

Q4: Harder to centrally manage permissions, security or backups across many devices.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.